MATHS MANTHAN
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e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 12** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, the angle between two consecutive ribs is: [1]

(A) $45°$ (B) $60°$ (C) $30°$ (D) $90°$

Q2. A horse is tied to a peg at one corner of a square grass field of side 15 m by means of a 5 m long rope. The area of the field the horse can graze is: (Use $\pi = 3.14$) [1]

(A) 19.625 m^2 (B) 78.5 m^2 (C) 25 m^2 (D) 39.25 m^2

Q3. A circular disc of radius 20 cm is divided into 4 concentric parts of equal area. The radius of the innermost circle is: [1]

(A) 5 cm (B) 10 cm (C) $10\sqrt{2}$ cm (D) $5\sqrt{2}$ cm

Q4. If the perimeter of a square is equal to the perimeter of a circle, then the ratio of the area of the square to the area of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) $\pi : 4$ (B) $4 : \pi$ (C) $14 : 11$ (D) $11 : 14$

Q5. A pendulum swings through an angle of $30°$ and describes an arc 8.8 cm in length. The length of the pendulum is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 16.8 cm (B) 14 cm (C) 21 cm (D) 18.2 cm

Q6. **Assertion (A):** If a wire of given length is bent first into a square and then into a circle, the area enclosed by the circle is greater than the area enclosed by the square.
Reason (R): For a fixed perimeter, the ratio of the area of a square to that of a circle is $\pi : 4 \approx 11 : 14 < 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, find the area between two consecutive ribs. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

Q8. The radius of a wheel is 0.25 m. Find the number of complete revolutions it will make to travel a distance of 11 km. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A square metal drain plate has side 28 cm. Inside the plate, 25 identical circular holes of diameter 2.8 cm each are drilled to drain water. Find the remaining area of the plate. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

Q10. An equilateral triangle has side 24 cm. From each vertex, an arc of radius 12 cm is drawn touching the other two arcs. Find the area of the bounded space between the arcs. (Use $\pi = 3.14$, $\sqrt{3} = 1.732$) [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. A circular municipal park of radius 21 m is surrounded by an outer jogging path 3.5 m wide. A protective steel fence is erected along the outer boundary of the path, and the entire path is graveled.

(a) Find the inner and outer radii and circumferences of the path.

(b) Calculate the surface area of the jogging path.

(c) If fencing costs 25 per metre and graveling costs 16 per m^2, calculate the total cost of the improvement project. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[5]

Q12. In a circle of radius 28 cm, two diameters AB and CD are perpendicular to each other. With OD as diameter, a smaller circle is drawn.

(a) Find the radius and area of the smaller circle.

(b) Find the area of $\triangle ABC$ formed by diameter AB and vertex C.

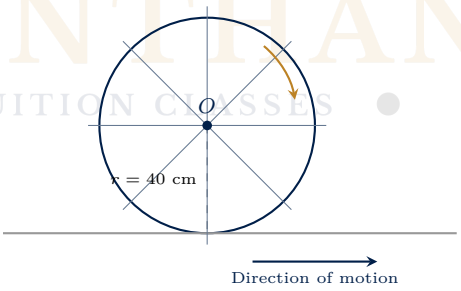
(c) Calculate the area of the shaded region consisting of the small circle and the two segments of the larger circle bounded by AC, BC and arc ACB. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Automotive Speedometer and Wheel Transmission Physics

A vehicle instrumentation engineer tests the odometer and speedometer accuracy of a car. The car wheels have an outer diameter of $d = 80$ cm. The vehicle is driven at a steady cruising speed of 66 km/h for a test duration of 10 minutes.



(i) Calculate the distance traveled by the car in one complete revolution of a wheel. [1]

(ii) Find the total distance traveled by the car in 10 minutes. [1]

(iii) Calculate the total number of complete revolutions each wheel makes in 10 minutes, OR find the number of revolutions per minute (RPM) of the wheels at this cruising speed. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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